BAHRAIN POLYTECHNIC بوليتكنك البحرين

Assessment Cover Sheet

Assessment Title	Project (Group)		
Assessment Type	Uncontrolled	Group	Not must-pass
Due Date	23 December 2026	**Course Code**	IT7006
Course Title	Object Oriented Design		
Internal Moderator's Name	Karim Ishtiaque		
External Examiner's Name			

Instructions:

1. This cover sheet must be completed (section in blue below) and attached to your assessment before submission into soft copy.
2. This carries a total of 100 marks distributed to a total of 11 tasks assessing CILO 1, 2 and 3.
3. The materials allowed for use in this assessment are EA, Visual Studio C# and Online resources permitted by the tutor.
4. This assessment covers 35% of the semester grade.
5. The **use of generative AI tools is strictly prohibited**.
6. References consulted (if any) must be properly acknowledged and cited.
7. The assessment has a total of 11 pages including the assessment cover sheet.

Learner ID		**Date Submitted**	
Learner Name			
Programme Code			
Programme Title			
Lecturer's Name			

By submiFng this assessment for marking, I affirm that this assessment is my own work.

Learner Signature	

Do not write beyond this line. For assessor use only.

Assessor's Name			
Marking Date		**Maks Obtained**	

Comments:

Policies:

- Requests for extensions should be made 48 hours (during working days) before the deadlines. The request must be sent by email to the Course Coordinator copying the class tutor and the Programming Program Manager. Extensions will only be approved for valid reasons. Extensions should be requested for the whole group.
- If an assessment is submitted late the maximum result the student can achieve is 60%.
- The cut off time for submitting an assessment will be 3 calendar days after the assessment is due.
- A student submitting after 3 calendar days will get 0%.
- The final grade for the assessment will be on an achievement basis only.
- There is **NO Resit** option for this assessment.

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Project Description

The project for this course will be to design and develop a windows application as a user / business layer using Enterprise Architect and C# programming language. The application should support a data layer. The application is expected to follow the object-oriented approach to demonstrate your understanding of and ability to use the concepts and topics discussed throughout the semester.

The project is designed for a business case with the expectation that it will include multiple functionalities and can be expanded into additional features as required.

Notes

- You are expected to work in groups of **6 (max)** students to produce a fully functioning application.
- Each member of the team is expected to do an equal amount of work and is expected to understand the entire application well enough to be examined on how the pieces fit together.
- The submission of the project should be made by one member of the group along with the work distribution plan for marking.
- Although the project focuses on object- oriented concepts, you may demonstrate additional design and development skills you acquired during the semester.

Team Formation and Dynamics

To form the group, meet in class with your classmates, exchange your schedules and ideas, upload a group formation form (Use the group formation template on Moodle). This project is designed to be challenging for a team of **six people for 20 to 24 class size and 5 for less than 20**. This means that if your team is not working well together, it will probably be reflected in the quality of your work. It is each team member's responsibility to make sure that everyone contributes their fair share of work. If you find that your team is experiencing a serious problem that puts the success of the project at risk, then you should immediately talk to your tutor. It is suggested that each member of the group handles an active role as follows:

Team Leader— the team, may find it useful to assign one person to manage the schedule and keep track of coordinating the different parts of the project. This person should mediate discussions about the project.

It is not the Team Leader's role to make all of the decisions or force their opinion on others, but if the team is split over a decision and cannot pick one or the other (even after enumerating benefits and negatives of each) it is the Team Leader's role to choose one of the options so that the project can move forward. The Team Leader should also hold the team members responsible for the tasks they are assigned.

It is the team leader's responsibility to schedule team meetings and with the tutor. Each student is required to participate during those meetings (*If possible*).

Recorder—this person records all decisions, tasks, and discussing points that take place in the meeting. It is their job to make sure that all meetings are documented to a point where any person doesn't complete a task; it is clear from the records that it was in fact assigned to them and that they should have done it by a given date. The recorder will take notes during meetings and create a thread in the group forum on Moodle.

Lead Integrator—this person takes on the role of managing the design and implementation side. They should be responsible for dealing with design or code conflicts. It is also their job to make sure that tasks are up to date and fully functioning. It is also this role's responsibility to stay on top of other team members' tasks and make sure that they complete their assignments. The integrator will ensure that the tutor is aware of the group member's active/passive participation

Time keeper - this person keeps track of the timeline of the project progress and keeps track of the meetings timings

Progress Check

In groups you may schedule or book a meeting (optional and will not be evaluated) with your tutor and individually present your work in 10 minutes. Each student may be asked to explain the work they have done themselves and work done by others as follows:

1. Part 1 Progress Check

You may meet with your tutor to go over material but questions of right or wrong are not allowed. You may sign up for a team meeting. An important goal of this meeting is to produce a schedule and deliverables for the second meeting. Bring your design document and project schedule / work distribution plan to the meeting. At the meeting you will be expected to be able to demonstrate a working model.

2. Part 2 Progress Check

The final submission will include the C# project – fully generated C# classes, a database file / source files with minimum 15 mock records in every table.

Deliverables

Each final project must demonstrate competency in the skills you have been learning and practicing in class.

Therefore, a significant portion of the final project grade comes from the following common specifications:

- ***Design Document:***
  - Design the following UML Diagram
    - **Use Case Diagram**. The diagram should demonstrate all your system functionalities.
    - **Class Diagrams**: The whole project is expected to have meaningful classes. Appropriate use of inheritance, association, abstract classes and class hierarchy is organized. Design demonstrates understanding of when to use "Is-a" vs. "Has-a".
    - **Sequence Diagram**: You are expected to create a diagram that depicts the interactions and sequence of messages exchanged between objects or components in a system over a specific period of time.
  - ERD: the full DB schema with all needed relations between your DB tables.
  - Design Pattern - You can select at least 2 design patterns (singleton cannot be the only design pattern)
  - The design document must be submitted as a **single Enterprise Architect (EA) project file containing all required diagrams**.

- ***User Interface Layer:***
  - Design input/output screens to show how your system functions. The screens should represent all the functional capabilities of your system. List all the reports that you believe will be helpful in addressing the information needs for this project. Design at least two of these reports.
  - C# must be used for the entire system. Source code must be well-commented. This includes comments for Classes, methods and internal (in-line) documentation for complex methods.
  - The submission should be a zip file that includes the project folder including the source code for your developed product using Visual Studio.

- ***Data Layer:***
  - The ERD should describe entities and their relationships. (3rd normal form). Entities must include Entity name, Keys (primary, foreign) and Attributes. Relationships should include Dependency, Association, and any Generalization.
  - The database should be designed and implemented to serve the functionality of the application.
    - The submission of your ERD must be as a local datafile that is included in your VS project.

Important Dates

For most recent dates refer to the course outline in Moodle

Week 03 – 27 Sept- 1 Oct	Teams submit project team list for approval (teams will be referred to as S1G1 (Stream 1 and Group 1)
Week 10 – 22-26 Nov 2026	Progress Check (Design Part) - **Optional**
Week 13 - 23 Dec 2026	Project Submission Date

Submissions

You are a consulting team that provides a solution information system. Your consulting group has been chosen to perform the task of designing business/information-processing solutions. Use information available from this description, design techniques you have learned and available tools to prepare your responses to each of the detailed requirements.

Your submission before due dates should follow:

- Project must be submitted via Moodle, zipped file name should be in the form: **Stream(X)_Group(X)**
  Example: S1G1
- The compressed file should contain the following:
  - Project Report
  - EA Project File
  - C# Project File
  - Demo Video File

Business case

Scenario:

Bahrain Institute of Technology (BIT) is launching a new **Smart Course Advising Information System (SCAIS)** for their flagship programs to assist faculty advisers in managing and guiding their advisees efficiently. The system will be used to recommend and validate course selections for each student based on their academic history, specialization, prerequisites and corequisites.

As part of this initiative, you have been hired as a junior developer to build a **Windows desktop application** using **C#.NET Framework 4.8 and SQL Server LocalDB ,** with future expansion planned for web and mobile platform.

Objective:

To design and implement a system that automates course advising, ensuring that IT students enroll in the correct subjects while adhering to prerequisite rules. It also provides insights for academic planning and tracking specialization progress.

Types of Users:

1. **Adviser** - Faculty member who manages advisees and approves course selections.
2. **Student(Advisee)** – IT Student with a declared specialization.
3. **Administrator** – Maintains the course catalog, user accounts, and system rules.

Functional Requirements for Each User

- **Adviser Features:**

- Log in and view assigned advisees
- Access academic history and specialization for each advisee
- Recommend courses based on eligibility (completed prerequisites/corequisites)
- Approve or modify course plans submitted by advisees
- Generate course progression reports

- **Student Features:**

- Log in and view academic record
- View list of eligible courses for the coming semester
- Submit preferred courses for adviser review
- View approved courses and remarks from the adviser

- **Administrator Features:**

- Manage user accounts (advisers and students)
- Add/update course catalog with prerequisites and requisites
- Assign advisees to advisers
- Define course offerings per semester
- Maintain specialization structures and curriculum flowcharts

Curriculum Structure Examples:

Each specialization **(Programming, Networking, Cybersecurity, Database)** has:

- **Core subjects (shared by all)**
  These are common subjects that all students must take, regardless of their chosen specialization
  Examples: Computer Systems, Math for Computing, Unix System etc.

- **Specialized Subjects (unique to track)**
  These are unique to each specialization and help students gain **in-depth knowledge** in their chosen track.
  Example:
  - For **Programming:** Object Oriented Design
  - For **Networking:** Wireless Communication
  - For **Cybersecurity:** Defense in Depth
  - For **Database:** Database Administration

- **Subject rules:**
  Some courses or subjects must be taken before others. These are called **prerequisites rules,** and they ensure that a student has the **necessary foundational knowledge** before moving on to more advanced topics.
  Example:
  - **Database 1** is a prerequisite for **Database 2**
  - **Advanced Networking** is a prerequisite for **Networking and Data Communications 2**

  In a course advising system, these rules are used to **automatically validate course eligibility.** If a student has not passed the required prerequisite, the system should **block** or **warn** them from enrolling in the dependent course.

- **Other Considerations/Requirements:**

Role-based login(Admin, Adviser, Student)

- The system must support different user roles, with access tailored to what each role needs to do

Single login screen

- All users(admin, adviser, student) log in through one unified screen. The system redirects them based on their role

Course validation logic

- The system must automatically check if a student meets **prerequisites** and **co-requisites** before allowing course enrollement

Track course status

- It should show a student's **completed, currently enrolled(in-progress),** and **yet to be taken(pending)** courses

Database design

The database should include the necessary tables like:

- Users
- Students, Advisers
- Courses – Course details
- Enrollments – which student enrolled in which course
- Specialization – Programming, Networking, Cybersecurity, Database
- Prerequisites – Rules linking course dependencies

- **Technology Stack:**
  - **CASE/UML/MDD:** Sparx Enterprise Architect
  - **Frontend:** C# , Windows Forms(.NET 4.8)
  - **Backend:** SQL Server LocalDB

Note:

The requirements can be expanded and assumptions can be made as the project may require.

Marking Criteria

For the implementation part marking will be visual.

If your project is not running a feature, your mark for that feature will be 0, but if it is running then marking will be based on quality of the code.

Part	Max Marks
A. Project Overview	10
B. UML Design	25
C. Object-Oriented Concepts	25
D. Design Pattern	20
E. User Interface & System Implementation	10
F. Data Layer & Database	10
TOTAL	**100**

Note: Detailed information for each part is provided in the Marking Rubric document

Important notes:

- For an 'A+' grade the student is expected to get above 95% of the subject mark.
- For an 'A' grade the student is expected to get between 90% and 95% of the subject mark.
- For an 'A-' grade the student is expected to get between 85% and 90% of the subject mark.
- For a 'B+' grade the student is expected to get between 80% and 85% of the subject mark.
- For a 'B' grade the student is expected to get between 75% and 80% of the subject mark.
- For a 'B-' grade the student is expected to get between 70% and 75% of the subject mark.
- For a 'C+' grade the student is expected to get between 65% and 70% of the subject mark.
- For a 'C' grade the student is expected to get between 60% and 65% of the subject mark.
- If a student gets under 60 then the student will get an 'F' grade and fail the subject.

Learning Outcomes

Upon the completion, this project will support the following learning outcomes of the course:

1. Demonstrate advanced knowledge of object-oriented architectural design and fundamental design principles, activities and environment.

2. Design and analyse Use Cases and apply Design Patterns at a specialist level.

3. Implement object-oriented user interfaces to databases using some specialist applications.

********* End of the Project *********